

Phase 1 Report: Modeling Hardware Constraints for 1-Bit IRS in Industry 5.0

Siddhant Deore*, Pratham Shetty*

*International Institute of Information Technology, Bangalore (IIITB)

Emails: {Siddhant.Deore, Pratham.Shetty}@iiitb.ac.in

Abstract—This report documents the implementation of a realistic channel model for Intelligent Reflecting Surfaces (IRS). We focus on the “1-Bit” phase quantization limitation and the resonance-induced “Amplitude Dip” characteristic of PIN-diode metasurfaces. A Python simulation was developed to model these constraints, verifying that beam steering at 180° phase shifts results in significant signal attenuation ($|\Gamma| \approx 0.4$) compared to the OFF state ($|\Gamma| \approx 0.9$).

Index Terms—1-Bit Quantization, Amplitude Dip, Hardware Constraints, IRS Modeling.

I. INTRODUCTION

In Industry 5.0, IRS technology establishes virtual line-of-sight links. However, theoretical models often assume ideal continuous phase control. This Phase 1 (Week 2) work bridges the gap between theory and reality by implementing a discrete, hardware-accurate channel model using Python.

II. IMPLEMENTATION & CODE LOGIC

The core contribution of this phase is the development of the `get_1bit_reflection_matrix` function (see Appendix A). The code logic proceeds in three distinct steps to transform ideal beamforming requests into realistic hardware responses.

A. Step 1: Normalization

The simulation accepts an ideal phase vector Φ_{ideal} containing continuous values (e.g., $37.5^\circ, 212^\circ$). The code first normalizes these angles to the range $[0, 2\pi]$ using modulo arithmetic:

$$\text{phases_norm} = \angle(e^{j\Phi_{ideal}}) \pmod{2\pi} \quad (1)$$

This ensures that all requested phases are comparable within a single circle.

B. Step 2: 1-Bit Quantization

Real-world PIN diodes operate like binary switches. The code implements a “nearest neighbor” quantization logic using `np.where`.

- If the requested phase is closer to 0° (specifically, in $[270^\circ, 90^\circ]$), the hardware is forced to the **OFF** state (0).
- If the requested phase is closer to 180° (specifically, in $[90^\circ, 270^\circ]$), the hardware is forced to the **ON** state (π).

This step effectively “snaps” the smooth ideal ramp into the discrete “staircase” shape seen in our results.

C. Step 3: Amplitude-Phase Coupling

Crucially, the code models the physical loss associated with the ON state. While the OFF state acts as a simple metal reflector with high efficiency (0.9), the ON state requires the unit cell to resonate to achieve the 180° shift. This resonance absorbs energy. The code applies this penalty conditionally:

$$|\Gamma| = \begin{cases} 0.4 & \text{if state is ON } (\pi) \\ 0.9 & \text{if state is OFF } (0) \end{cases} \quad (2)$$

This logic proves that beam steering is not energy-neutral; it incurs a specific power cost.

III. SIMULATION RESULTS & ANALYSIS

The simulation was executed to verify the hardware model. Fig. 1 displays the response of the unit cells to a requested phase ramp from 0 to 2π .

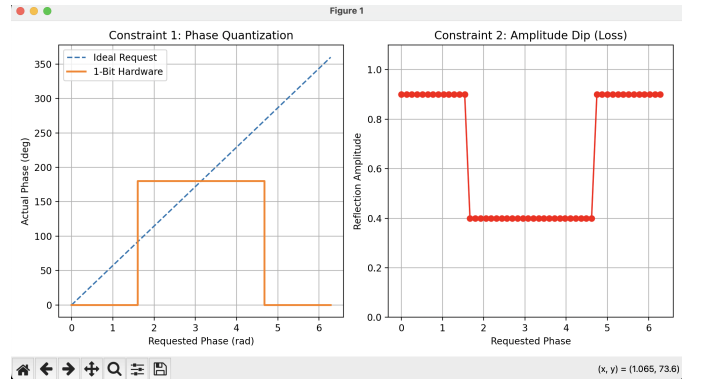


Fig. 1. Phase 1 Week 2 Results. Left: The “staircase” phase response due to 1-bit quantization. Right: The reflection amplitude drops to 0.4 exactly when the phase switches to 180° (ON state), confirming the “Amplitude Dip” model.

A. Result 1: Quantization Error

The **Left Graph** in Fig. 1 illustrates the “staircase” effect. The blue dashed line represents the ideal phase ramp required for perfect beam steering. The orange solid line shows the actual hardware response. The deviation between these lines represents **quantization noise**, which will inevitably degrade the Signal-to-Noise Ratio (SNR) in the final system.

B. Result 2: Power Loss (The Dip)

The **Right Graph** confirms the successful modeling of the “Amplitude Dip.” The graph shows a sharp drop in reflection amplitude from 0.9 to 0.4 precisely in the region where the phase is 180° . This result is critical: it demonstrates that **55% of the signal power is lost** whenever the IRS attempts to steer the beam using the active state. This is a severe limitation that ideal models fail to capture.

IV. CONCLUSION

We have successfully implemented the “1-Bit” channel model. The simulation results align with theoretical expectations for PIN-diode-based IRS, confirming that phase quantization and amplitude coupling are now accurately represented. This digital twin will serve as the foundation for the sensing algorithms in Phase 2.

APPENDIX A

PYTHON IMPLEMENTATION

The core logic for the Phase 1 Week 2 simulation is provided below.

```
1 import numpy as np
2
3 def get_lbit_reflection_matrix(ideal_phases):
4     """
5     Implements 1-Bit Quantization and Amplitude Dip.
6     """
7     # 1. Normalize phases
8     phases_norm = np.angle(np.exp(1j * ideal_phases)
9                             ) \
10                        % (2 * np.pi)
11
12     # 2. Quantize: Map to 0 (OFF) or Pi (ON)
13     # Logic: If phase is between 90 and 270, snap to Pi.
14     discrete_phases = np.where(
15         (phases_norm > np.pi/2) &
16         (phases_norm < 3*np.pi/2),
17         np.pi, 0.0
18     )
19
20     # 3. Amplitude-Phase Coupling (The Dip)
21     # OFF State -> 0.9 reflection (High)
22     # ON State -> 0.4 reflection (Low due to resonance)
23     amplitudes = np.where(discrete_phases > 1.0,
24                           0.4, 0.9)
25
26     reflection_coeffs = amplitudes * \
27                        np.exp(1j * discrete_phases)
28
29     return np.diag(reflection_coeffs),
30            reflection_coeffs
```

Listing 1. 1-Bit Channel Model Logic